



**Dandes
Academy**

 IMDB Movies Database – SQL & Data Modeling Assignment

Dandes Academy – SQL & Data Modeling Practice

**Author
Srinivas Dande**





IMDB Movies Database – SQL & Data Modeling Assignment

1. Overview
2. What You Will Learn – Skill Coverage
3. Tables List - 6 Tables
4. Relationships & Cardinality
5. Table Creation Script
6. Data Load Section
7. Assignment Questions



1. Overview

- IMDB Movies SQL & Data Modeling Assignment – Structured Across 8 SQL Skill Areas
- This IMDB Movies Database is designed as a complete SQL practice environment, covering every important SQL concept required for:
 - ✓ Job interviews
 - ✓ Real-world data engineering
 - ✓ Data analytics
 - ✓ Business intelligence
 - ✓ Data modeling
- The assignment contains 75 questions, divided into 8 deep-dive sections, each focusing on a core SQL skill area.

2. What You Will Learn – Skill Coverage

- Below are the 8 SQL sections.

Section 1: Data Exploration & Data Quality

Section 2: Date, Time & Country Trends

Section 3: Genre Insights & Aggregations

Section 4: Movie Ratings & Title Analysis

Section 5: Production House Analytics

Section 6: People Analytics (Directors, Actors, Actresses)

Section 7: SQL Performance & Index Optimization

Section 8: Business Insights & Interpretation



Section 1: Data Exploration & Data Quality

- ✓ Understanding table shapes
- ✓ Checking NULLs & missing values
- ✓ Basic filtering
- ✓ Distinct values
- ✓ Data completeness & profiling

Section 2: Date, Time & Country Trends

- ✓ Extracting year, month, day from dates
- ✓ Year-wise & month-wise trends
- ✓ Country-wise production patterns
- ✓ Range queries using dates

Section 3: Genre Insights & Aggregations

- ✓ Genre-level analysis
- ✓ GROUP BY, HAVING, COUNT, AVG
- ✓ Window functions (running totals, moving averages)
- ✓ Ranking genres
- ✓ Categorizing movies using CASE

Section 4: Movie Ratings & Title Analysis

- ✓ Ranking movies (avg/median ratings)
- ✓ Comparing mean vs median
- ✓ Title-pattern search
- ✓ Year-wise best movies
- ✓ Above-average movie analysis



Section 5: Production House Analytics

- ✓ Ranking production houses
- ✓ Revenue-based analysis
- ✓ Hit movie performance (avg_rating > 8)
- ✓ Genre-wise strengths
- ✓ Identifying consistently strong companies

Section 6: People Analytics (Directors, Actors, Actresses)

- ✓ Best actors/actresses (ratings + votes)
- ✓ Top directors by ratings
- ✓ Weighted averages using vote counts
- ✓ Multi-genre actors
- ✓ Actor-director collaboration counts
- ✓ Director career timeline analysis

Section 7: SQL Performance & Index Optimization

- ✓ Using EXPLAIN
- ✓ Designing correct indexes
- ✓ Rewriting slow queries
- ✓ Function-based indexing
- ✓ Performance-aware JOIN strategies

Section 8: Business Insights & Interpretation

- ✓ Converting SQL data → business decisions
- ✓ Genre profitability insights
- ✓ Market-level patterns (country, year)
- ✓ Ratings vs revenue correlation
- ✓ Language, duration & audience response



These 8 sections help you master:

- SQL fundamentals
- Advanced SQL analytics
- Window functions
- Ranking & classification
- JOINS & multi-table queries
- Real-world business insights
- Performance tuning

3. Tables List - 6 Tables

- **IMDB Movies Database contains 6 schema tables:**
 - 1) movie – Movie metadata (title, year, languages, income, company...)
 - 2) genre – Genre mappings per movie
 - 3) ratings – Rating details (avg, median, total_votes)
 - 4) names – People (actors, actresses, directors)
 - 5) director_mapping – Movie ↔ Director link
 - 6) role_mapping – Movie ↔ Actor/Actress/Role link
- **This structure is inspired by IMDB but optimized for SQL learning.**



4. Relationships & Cardinality

A) Movie → Genre (1 : Many)

- ✓ A movie can have multiple genres.
- ✓ Each record in GENRE = (movie_id + genre).
- ✓ (1 Movie → Many Genre rows)

B) Movie → Ratings (1 : 1)

- ✓ Each movie has exactly one ratings entry.
- ✓ ratings.movie_id is PK + FK.
- ✓ (1 Movie → 1 Ratings row)

C) Movie ↔ Names (Directors) (Many : Many)

- ✓ A movie can have multiple directors.
- ✓ A director can direct many movies.
- ✓ Implemented using director_mapping.
- ✓ (Many Movies ↔ Many Directors)

D) Movie ↔ Names (Cast Roles) (Many : Many)

- ✓ A movie can have many actors.
- ✓ An actor can perform in many movies.
- ✓ role_mapping links movies ↔ names with category.
- ✓ (Many Movies ↔ Many Actors)

E) Names Table (Master Table)

- ✓ Stores all people (actors, actresses, directors, etc).
- ✓ Referenced by both director_mapping and role_mapping.
- ✓ Independent entity; no dependency on movie.



5. Table Creation Script

-- Creating and using the database

```
DROP DATABASE IF EXISTS myda_imdb;
```

```
CREATE DATABASE myda_imdb;
```

```
USE myda_imdb;
```

-- Creating the structure of various tables in the database

-- 1. MOVIE MASTER

```
-- One row per movie
```

```
-----
```

```
DROP TABLE IF EXISTS movie;
```

```
CREATE TABLE movie (
```

```
  id          VARCHAR(10) NOT NULL, -- Movie ID (primary key from dataset)
```

```
  title       VARCHAR(200) NOT NULL, -- Movie title
```

```
  year        INT,              -- Release year
```

```
  date_published DATE,          -- Exact release date
```

```
  duration    INT,              -- Duration in minutes
```

```
  country     VARCHAR(250),      -- Comma-separated countries
```

```
  worldwide_gross_income VARCHAR(30), -- As text from dataset (keep as-is)
```

```
  languages   VARCHAR(200),      -- Comma-separated languages
```

```
  production_company VARCHAR(200) -- Name of production company
```

```
);
```




-- 2. GENRE

-- A movie can have multiple genres (1-to-many from movie)

DROP TABLE IF EXISTS genre;

```
CREATE TABLE genre (  
    movie_id VARCHAR(10),  
    genre VARCHAR(20)
```

```
);
```

-- 3. NAMES

-- People: actors, actresses, directors, etc.

DROP TABLE IF EXISTS names;

```
CREATE TABLE names (  
    id          VARCHAR(10) NOT NULL, -- Person ID (from dataset)  
    name        VARCHAR(100),         -- Full name  
    height      INT,                  -- Height in cm (may be NULL)  
    date_of_birth DATE,               -- DOB (may be NULL)  
    known_for_movies VARCHAR(100)     -- Comma-separated movie IDs (as in dataset)  
);
```

-- 4. DIRECTOR MAPPING

-- Many-to-many : movie <-> director (name)

DROP TABLE IF EXISTS director_mapping;

```
CREATE TABLE director_mapping (  
    movie_id VARCHAR(10), -- References movie(id)  
    name_id VARCHAR(10)  -- References names(id)  
);
```



-- 5. ROLE MAPPING

-- Many-to-many : movie <-> cast/crew
-- category = 'actor', 'actress', etc.

DROP TABLE IF EXISTS role_mapping;

```
CREATE TABLE role_mapping (  
    movie_id VARCHAR(10), -- References movie(id)  
    name_id VARCHAR(10), -- References names(id)  
    category VARCHAR(10)      -- actor / actress / director / etc.  
);
```

-- 6. RATINGS

-- One ratings row per movie

DROP TABLE IF EXISTS ratings;

```
CREATE TABLE ratings (  
    movie_id VARCHAR(10) NOT NULL, -- References movie(id)  
    avg_rating DECIMAL(3,1) NOT NULL, -- Average rating (e.g., 7.5)  
    total_votes INT, -- Number of votes  
    median_rating INT -- Median rating (1-10)  
);
```

6. Data Load Section

- **Open IMDB DataSet Import.sql**
 - ✓ You can see all table creation statements
 - ✓ You can see row insert statements
- **Run the Script**
 - ✓ So that All tables and rows will be ready



7. Assignment Questions

- This assignment has 8 sections.
- Questions for each section are given in a separate SQL file.
- You must write your SQL code directly below each question in the corresponding SQL file.

- **Section 1 - Data Exploration & Data Quality.sql**
 - This Section has 10 Questions from Q1 to Q10
- **Section 2 - Date, Time & Country Trends.sql**
 - This Section has 8 Questions from Q11 to Q18
- **Section 3 - Genre Insights & Aggregations.sql**
 - This Section has 12 Questions from Q19 to Q30
- **Section 4 - Movie Ratings & Title Insights.sql**
 - This Section has 8 Questions from Q31 to Q38
- **Section 5 - Production House Analytics.sql**
 - This Section has 10 Questions from Q39 to Q48
- **Section 6 - People Analytics – Directors, Actors, Actresses.sql**
 - This Section has 12 Questions from Q49 to Q60
- **Section 7 - SQL Performance & Index Optimization.sql**
 - This Section has 5 Questions from Q61 to Q65
- **Section 8 - Business Interpretation & Insights.sql**
 - This Section has 10 Questions from Q66 to Q75